

• It should be noted that the number of minterms appearing in the normal form of a formula is the same as the number of Tentries in the column corresponding to the formula.

Example 3: Obtain principal disjunctive normal form for $p \vee \neg q$.

$$\begin{aligned}
 p \vee \neg q &\equiv (p \wedge T) \vee (\neg q \wedge T) \quad (1st \text{ Identity Law}) \\
 &\equiv (p \wedge (q \vee \neg q)) \vee (\neg q \wedge (p \vee \neg p)) \quad (1st \text{ Negation Law}) \\
 &\equiv ((p \wedge q) \vee (p \wedge \neg q)) \vee ((\neg q \wedge p) \vee (\neg q \wedge \neg p)) \quad (2nd \text{ Distributive Law}) \\
 &\equiv ((p \wedge q) \vee (p \wedge \neg q)) \vee ((p \wedge \neg q) \vee (\neg p \wedge \neg q)) \quad (2nd \text{ Commutative Law}) \\
 &\equiv (p \wedge q) \vee ((p \wedge \neg q) \vee (\neg p \wedge \neg q)) \quad (1st \text{ Associative Law}) \\
 &\equiv (p \wedge q) \vee ((p \wedge \neg q) \vee (\neg p \wedge \neg q)) \quad (1st \text{ Associative Law}) \\
 &\equiv (p \wedge q) \vee ((p \wedge \neg q) \vee (\neg p \wedge \neg q)) \quad (1st \text{ Idempotent Law}) \\
 &\equiv (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q) \quad (1st \text{ Associative Law})
 \end{aligned}$$

Example 4: Obtain principal disjunctive normal form for $(p \wedge q) \vee (\neg p \wedge \neg q)$.

$$\begin{aligned}
 (p \wedge q) \vee (\neg p \wedge \neg q) &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) \quad (2nd \text{ Identity Law}) \\
 &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) \quad (2nd \text{ Negation Law}) \\
 &\equiv (p \wedge q)
 \end{aligned}$$

$$\begin{aligned}
 p \wedge q &\equiv (p \wedge q) \wedge T \quad (1st \text{ Identity Law}) \\
 &\equiv (p \wedge q) \wedge (p \vee \neg p) \quad (1st \text{ Negation Law}) \\
 &\equiv ((p \wedge q) \wedge p) \vee ((p \wedge q) \wedge \neg p) \quad (2nd \text{ Distributive Law}) \\
 &\equiv (p \wedge q \wedge p) \vee (p \wedge q \wedge \neg p)
 \end{aligned}$$

$$\begin{aligned}
 \neg p \wedge \neg q &\equiv (\neg p \wedge \neg q) \wedge T \quad (1st \text{ Identity Law}) \\
 &\equiv (\neg p \wedge \neg q) \wedge (q \vee \neg q) \quad (1st \text{ Negation Law}) \\
 &\equiv ((\neg p \wedge \neg q) \wedge q) \vee ((\neg p \wedge \neg q) \wedge \neg q) \quad (2nd \text{ Distributive Law}) \\
 &\equiv (\neg p \wedge \neg q \wedge q) \vee (\neg p \wedge \neg q \wedge \neg q)
 \end{aligned}$$

$$\begin{aligned}
 q \wedge p &\equiv (q \wedge p) \wedge T \quad (1st \text{ Identity Law}) \\
 &\equiv (q \wedge p) \wedge (p \vee \neg p) \quad (1st \text{ Negation Law}) \\
 &\equiv ((q \wedge p) \wedge p) \vee ((q \wedge p) \wedge \neg p) \quad (2nd \text{ Distributive Law}) \\
 &\equiv (q \wedge p \wedge p) \vee (q \wedge p \wedge \neg p)
 \end{aligned}$$